

# Lecture 8: Monte Carlo Methods

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# Outline

- 1 Random Variable Generation
- 2 Monte Carlo Integration
- 3 Asymptotic Analysis: Law of Large Numbers
- 4 Non-asymptotic Analysis: Inequalities

# History

- Basic Monte Carlo methods: formally proposed by Stanislaw Ulam & John Von Neumann in 1940s at Los Alamos National Lab (Named after a casino in Monaco)
- Markov Chain Monte Carlo methods: formally proposed by Metropolis et al in 1950s at Los Alamos National Lab

# Widely Applications

Monte Carlo methods have been used in various tasks, including

- Simulating a system and sampling from the underlying probability distribution  $f(x)$
- Estimating a quantity through Monte Carlo integration

$$c = E_{\pi}(h(x)) = \int f(x)h(x)dx;$$

- Optimizing a target function to find its maxima or minima
- Learning parameters from a training set to optimize some loss functions



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- 4 Non-asymptotic Analysis: Inequalities

# Sampling

- Assuming an algorithm is available for generating Unif(0, 1) random numbers
- Two elementary methods for generating random variables
  - ▶ Inverse-transform method: operates on the CDF
  - ▶ The acceptance-rejection method: operates on the PDF (or PMF)

# Inverse Transform Method

- Given a  $\text{Unif}(0, 1)$  r.v., we can construct an r.v. with any continuous distribution we want.
- Conversely, given an r.v. with an arbitrary continuous distribution, we can create a  $\text{Unif}(0, 1)$  r.v.
- Other names:
  - ▶ probability integral transform
  - ▶ inverse transform sampling
  - ▶ the quantile transformation
  - ▶ the fundamental theorem of simulation

# Inverse Transform Method: Recall

## Theorem

Let  $F$  be a CDF which is a continuous function and strictly increasing on the support of the distribution. This ensures that the inverse function  $F^{-1}$  exists, as a function from  $(0, 1)$  to  $\mathbb{R}$ . We then have the following results.

- 1 Let  $U \sim \text{Unif}(0, 1)$  and  $X = F^{-1}(U)$ . Then  $X$  is an r.v. with CDF  $F$ .
- 2 Let  $X$  be an r.v. with CDF  $F$ . Then  $F(X) \sim \text{Unif}(0, 1)$ .

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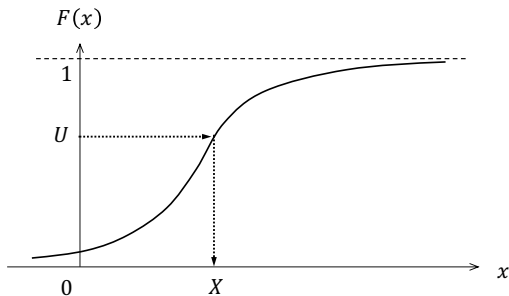
## Algorithm Inverse-Transform Method: PDF Case

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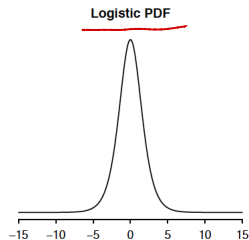
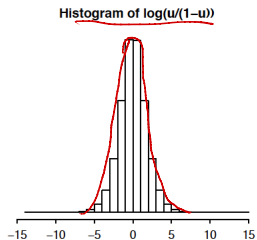
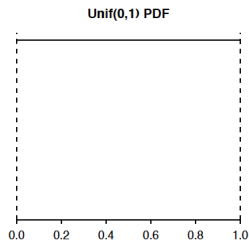
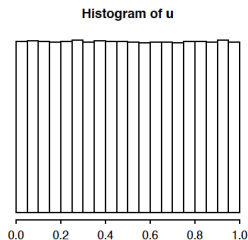
**input:** Cumulative distribution function  $F$ .

**output:** Random variable  $X$  distributed according to  $F$ .

- 1: Generate  $U$  from  $Unif(0, 1)$ .
  - 2:  $X \leftarrow F^{-1}(U)$
  - 3: **return**  $X$
- 



# Histogram & PDF: Example



# Box-Muller: Recall

Let  $U \sim \text{Unif}(0, 2\pi)$ , and let  $T \sim \text{Expo}(1)$  be independent of  $U$ . Define  $X = \sqrt{2T} \cos U$  and  $Y = \sqrt{2T} \sin U$ . Find the joint PDF of  $(X, Y)$ . Are they independent? What are their marginal distributions?

# Box-Muller Method

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**Algorithm** Normal Random Variable Generation: Box-Muller Approach

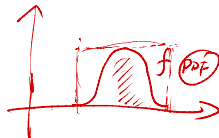
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**output:** Independent standard normal random variables  $X$  and  $Y$ .

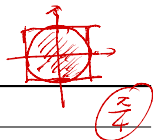
- 1: Generate two independent random variables,  $U_1$  and  $U_2$ , from  $Unif(0, 1)$ .
  - 2:  $X \leftarrow (-2 \ln U_1)^{1/2} \cos(2\pi U_2)$
  - 3:  $Y \leftarrow (-2 \ln U_1)^{1/2} \sin(2\pi U_2)$
  - 4: **return**  $X, Y$
-



# Acceptance-Rejection Method



- Suppose one can generate samples (relatively easily) from PDF  $g$
- How can random samples be simulated from PDF  $f$ ?



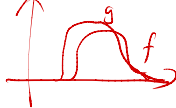
## Algorithm Acceptance-Rejection Algorithm

Let  $c$  denote a constant such that  $c \geq \sup_y \frac{f(y)}{g(y)}$ . Then:

Step 1: Generate  $Y \sim g$ .

Step 2: Generate  $U \sim \text{Unif}[0, 1]$ .

Step 3: If  $U \leq \frac{f(Y)}{c \cdot g(Y)}$ , set  $X = Y$ . Otherwise go back to step 1.



# Acceptance-Rejection Method

event  $A = "U \leq \frac{f(Y)}{c \cdot g(Y)}"$

(i) the PDF of generated r.v.s.  $f_Y(y|A) = \frac{P(A|Y=y)}{P(A)} \cdot f_Y(y)$

$$\textcircled{1} P(A|Y=y) = P\left(U \leq \frac{f(Y)}{c \cdot g(Y)} \mid Y=y\right) = P\left(U \leq \frac{f(y)}{c \cdot g(y)} \mid Y=y\right)$$

## Theorem

(i) The random variable generated by the Acceptance-Rejection method has the desired PDF  $f$ .

(ii) The number of iterations of the algorithm that are needed is a first-success random variable with mean  $c$ .

(iii)  $c \geq 1$  unif(0,1) ①

$$= P\left(U \leq \frac{f(y)}{c \cdot g(y)}\right) = \frac{f(y)}{c \cdot g(y)}$$

$$P(U \leq x) = x, \quad 0 \leq x \leq 1 \quad \text{CDF of Unif(0,1)}$$

## Proof

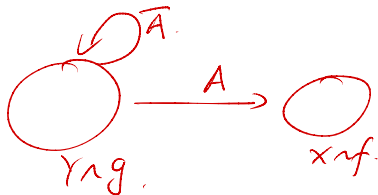
$$\begin{aligned} \textcircled{2} \quad P(A) &\stackrel{\text{LoTP}}{=} \int \underbrace{P(A|Y=y)}_{c \cdot g(y)} \cdot g(y) dy = \int \frac{f(y)}{c \cdot g(y)} \cdot g(y) dy \\ &= \int \frac{f(y)}{c} dy = \frac{1}{c} \int f(y) dy = \frac{1}{c} \leq 1 \quad (c \geq 1) \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad f_Y(y|A) &= \frac{P(A|Y=y)}{P(A)} \cdot \underbrace{f_Y(y)}_{f(y)} \\ &= \frac{\frac{f(y)}{c \cdot g(y)}}{\frac{1}{c}} \cdot g(y) = \underline{f(y)}, \text{ by } \end{aligned}$$

$$\textcircled{X} \sim \textcircled{f}$$

# Proof

(ii)



# of iterations  $Z \sim \text{FSCP}$

$$P = P(A) = \frac{1}{c} \leq 1.$$

$$\Rightarrow E(Z) = \frac{1}{P} = c$$

# Example: Beta Distribution

①  $X \sim \text{Beta}(2, 4)$ . PDF of  $X$ :  $f(x) = 20x(1-x)^3$ ,  $0 < x < 1$ .

② Acceptance-Rejection.  $Y \sim \text{unif}(0, 1)$ ,  $g(y) = 1$ .

$$C \geq \sup_y \frac{f(y)}{g(y)} = \sup_y 20y(1-y)^3 \Rightarrow y^* = \frac{1}{4}; \quad 0 < x < 1.$$

- Use the Acceptance-Rejection Method to generate a random variable with distribution Beta(2, 4)

$$\Rightarrow C \geq \frac{135}{64}$$

$$\text{pick } C = \frac{135}{64}$$

$$\Rightarrow \frac{f(y)}{C \cdot g(y)} = \frac{20y(1-y)^3}{\frac{135}{64} \cdot 1} = \frac{256}{27} y(1-y)^3, \quad 0 < y < 1$$

step 1: Generate  $Y \sim \text{unif}(0, 1)$

step 2:  $U \sim \text{unif}(0, 1)$

3: If  $U \leq \frac{f(Y)}{C \cdot g(Y)} = \frac{256}{27} Y(1-Y)^3$ , set  $X = Y$ .

otherwise

# Solution

# Example: Normal Distribution

①  $Z \sim N(0, 1)$   $(-\infty, +\infty)$

②  $X = |Z|$   $(0, +\infty)$

proposal distribution  
 $\frac{\text{Expo}(1)}{\text{Expo}(1)}$   $(0, +\infty)$

if we obtain  $X$ ,  $Z = \begin{cases} X & \text{w.p. } 0.5 \\ -X & \text{w.p. } 0.5 \end{cases}$   $Z \sim N(0, 1)$

- Use the Acceptance-Rejection Method to generate a random variable with distribution  $N(0, 1)$

③  $X = |Z|$ ;  $f_X(x) = \frac{2}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ ,  $0 < x < \infty$ .

choose  $g \sim \text{Expo}(1)$ ,  $g(x) = e^{-x}$ ,  $0 < x < \infty$ .

Now  $\frac{f(x)}{g(x)} = \frac{\frac{2}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}}{e^{-x}} = \frac{2}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2 + x}$   $C \geq \sup_x \frac{f(x)}{g(x)} = \frac{f(x^*)}{g(x^*)} = \frac{f(1)}{g(1)} = \sqrt{\frac{2e}{\pi}}$   $x^* = 1$

choose  $C = \sqrt{\frac{2e}{\pi}}$   $\Rightarrow \left( \frac{f(x)}{C \cdot g(x)} \right) = \exp\left\{-\frac{1}{2}(x-1)^2\right\}$

# Solution

④

Step 1: Generate  $Y \sim \text{Expo}(1)$

2:  $U \sim \text{unif}(0,1)$

3: If  $U \leq \exp\{-\frac{1}{2}(Y-1)^2\}$ , set  $X=Y$  Accept the sample

Otherwise, reject it, return to step 1.

4:  $U' \sim \text{unif}(0,1)$

$$Z' = \begin{cases} X & \text{if } U' \leq \frac{1}{2} \\ -X & \text{if } U' > \frac{1}{2} \end{cases}$$

$\Rightarrow \underline{Z \sim N(0,1)}$  | Compare  
Box-Muller.



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# Monte Carlo Integration

- We want to compute the integral of a function in a very high dimensional space  $\Omega$ :

$$c = \underbrace{E}_{\int}(\underbrace{h(\mathbf{x})}) = \int \underbrace{f(\mathbf{x})h(\mathbf{x})}d\mathbf{x};$$

- This is often estimated by Monte Carlo integration. By drawing  $n$  samples (empirical samples) from  $f(\mathbf{x})$ :

$$\underbrace{x_1, x_2, \dots, x_n} \sim \underbrace{f(\mathbf{x})} \quad \underbrace{h(x_1)} \dots \underbrace{h(x_n)}$$

- One can estimate  $c$  by the sample mean:

$$\frac{1}{n} \sum_{j=1}^n h(\mathbf{x}_j)$$

# Example: $\pi$ as An Integration

Evaluate the integration

$$\int_0^1 \frac{4}{1+x^2} dx.$$

$$\int_0^1 \frac{4}{1+x^2} \cdot 1 dx$$
$$= E_f[h(x)]$$

- $h(x) = 4/(1+x^2)$ ,  $f(x) = 1$ ,  $0 < x < 1$ .
- $X_1, \dots, X_n$ : samples from  $\text{Unif}(0, 1)$ .
- Monte-Carlo Integration:

$$h(x_i) = \frac{4}{1+x_i^2}$$

$$\int_0^1 \frac{4}{1+x^2} dx \approx \frac{1}{n} \sum_{i=1}^n \frac{4}{1+X_i^2}.$$

# Useful Tools: Importance Sampling

- Standard Monte Carlo integration is great if you can sample from the target distribution (i.e. the desired distribution)
- But what if you can't sample from the target?
- **Importance Sampling**: draw the sample from a proposal distribution and re-weight the integral using importance weights so that the correct distribution is targeted

# Importance Sampling

$$I = \int \underline{h(y)f(y)dy} \quad \textcolor{red}{E_f[h(Y)]}$$

- $h$  is some function and  $f$  is the PDF of random variable  $Y$
- When the PDF  $f$  is difficult to sample from, importance sampling can be used
- Rather than sampling from  $f$ , you specify a different PDF  $g$ , as the proposal distribution.

$$I = \int \underline{h(y)f(y)dy} = \int h(y) \frac{\textcolor{red}{f(y)}}{\textcolor{red}{g(y)}} g(y) dy = \int \frac{\textcolor{red}{h(y)f(y)}}{\textcolor{red}{g(y)}} \underline{g(y)dy}$$

# Importance Sampling

$$I = \underline{E_f[h(Y)]} = \int \frac{h(y)f(y)}{g(y)} g(y) dy = \underline{E_g \left[ \frac{h(Y)f(Y)}{g(Y)} \right]}$$

$h(y) \cdot \frac{f(y)}{g(y)}$   
weight.

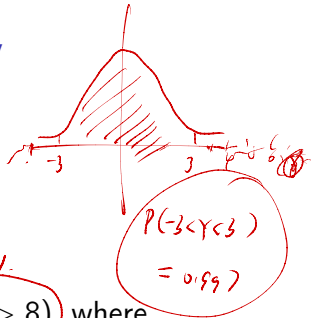
- Hence, given an iid sample  $Y_1, \dots, Y_n$  from PDF  $g$ , our estimator of  $I$  becomes

$$\hat{I} = \frac{1}{n} \sum_{j=1}^n \frac{h(Y_j)f(Y_j)}{g(Y_j)}$$

# Example: Gaussian Tail Probability

Key: Probability  $\longleftrightarrow$  Indicator Variable  $\longleftrightarrow$  Expectation

$$I(A) = \begin{cases} 1 & \text{if } A \text{ occurs.} \\ 0 & \text{otherwise} \end{cases}$$



$$C = P(Y > 8) = E[I(Y > 8)] = \int I(Y > 8) f(y) dy.$$

Evaluate the probability of rare event  $c = \mathbb{P}(Y > 8)$ , where  $Y \sim N(0, 1)$ .

$f \sim N(0, 1)$ ,  $h(y) = I(y > 8)$ .

①

$$\hat{C} = \frac{1}{n} \sum_{j=1}^n I(Y_j > 8)$$

$$Y_1, \dots, Y_n \sim N(0, 1)$$

$\neq 0$ .

②

$g \sim N(8, 1)$

# Solution

② Importance Sampling : choose  $g \sim N(8, 1)$ .

$$Y_1, \dots, Y_n \sim g(N(8, 1)).$$

$$\hat{C} = \frac{1}{n} \sum_{j=1}^n \frac{h(Y_j) \underbrace{f(Y_j)}_{g(Y_j)}}{g(Y_j)} = \frac{1}{n} \sum_{j=1}^n I(Y_j > 8) \cdot \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} Y_j^2}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} (Y_j - 8)^2}}$$

$$= \frac{1}{n} \sum_{j=1}^n I(Y_j > 8) \cdot e^{-8Y_j + 32}$$

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$$6.25 \times 10^{-16}$$



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Sample Mean: Recall  $\textcircled{1} E(\bar{X}_n) = \frac{1}{n} \sum_{j=1}^n E(X_j) = \frac{1}{n} \cdot n \mu = \mu$

unbiased estimation

$n$  samples from the same distribution

## Definition

Let  $X_1, \dots, X_n$  be i.i.d. random variables with finite mean  $\mu$  and finite variance  $\sigma^2$ . The sample mean  $\bar{X}_n$  is defined as follows:

$$\bar{X}_n = \frac{1}{n} \sum_{j=1}^n X_j$$

The sample mean  $\bar{X}_n$  is itself an r.v. with mean  $\mu$  and variance  $\sigma^2/n$ .

$$\textcircled{2} \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{j=1}^n \text{Var}(X_j) = \frac{1}{n^2} \cdot n \sigma^2 = \frac{1}{n} \sigma^2$$

$$\textcircled{n \rightarrow \infty}$$

$$\Rightarrow \text{Var}(\bar{X}_n) \rightarrow 0$$

$$\bar{X}_n \rightarrow \textcircled{\mu}$$

# Strong Law of Large Numbers (SLLN)

$$\frac{1}{n} (X_1 + \dots + X_n) \xrightarrow[n \rightarrow \infty]{\text{w.p. 1.}} E[X] = \mu.$$

## Theorem

The sample mean  $\bar{X}_n$  converges to the true mean  $\mu$  pointwise as  $n \rightarrow \infty$ , with probability 1. In other words, the event  $\bar{X}_n \rightarrow \mu$  has probability 1.

$$\frac{1}{n} (f(X_1) + \dots + f(X_n)) \xrightarrow[n \rightarrow \infty]{\text{w.p. 1.}} E[f(X)]$$

$$X_1, \dots, X_n, X \sim F.$$

# Weak Law of Large Numbers (WLLN)

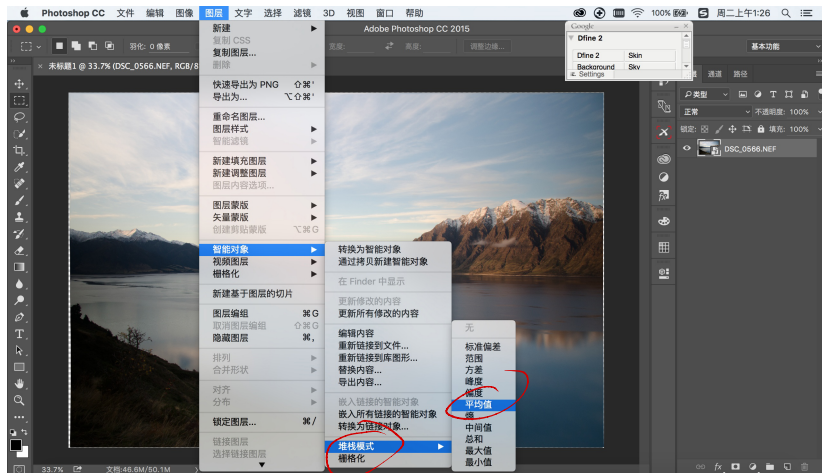
## Theorem

*For all  $\epsilon > 0$ ,  $P(|\bar{X}_n - \mu| > \epsilon) \rightarrow 0$  as  $n \rightarrow \infty$ . (This form of convergence is called convergence in probability).*

# Widely Applications: Photo Stacking with PC



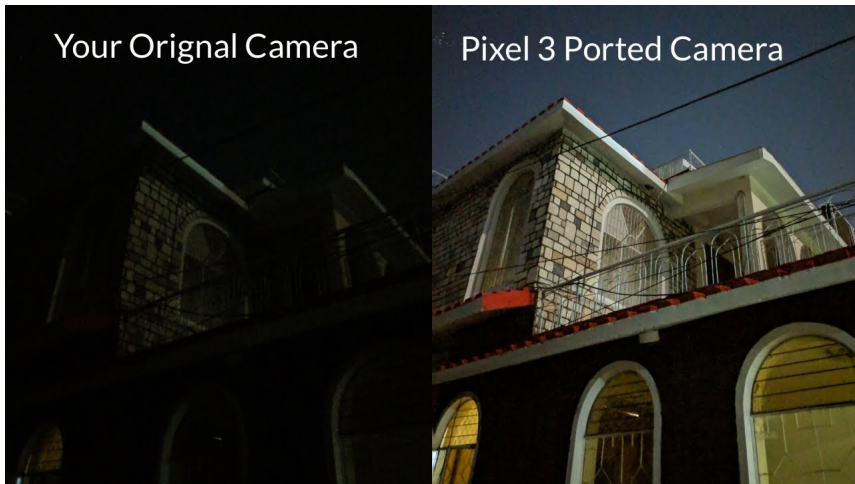
# Widely Applications: Photo Stacking with PC



# Widely Applications: Night Model with Smart Phone



# Widely Applications: Photo Stacking with Smart Phone





# Widely Applications: Photo Stacking with Smart Phone



# Widely Applications: Photo Stacking with Smart Phone



# Outline

1 Random Variable Generation

2 Monte Carlo Integration

3 Asymptotic Analysis: Law of Large Numbers

*Infinite Samples.*

4 Non-asymptotic Analysis: Inequalities

*Finite Samples.*

$$\mathbb{P}(|\bar{x}_n - \mu| > \varepsilon) \leq \alpha$$

# Cauchy-Schwarz Inequality: Recall

$\rho \in [-1, 1]$

Correlation

## Theorem

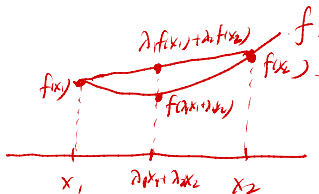
For any r.v.s  $X$  and  $Y$  with finite variances,

$$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}.$$

# Jensen's Inequality

$(x^2)$

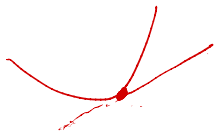
$$f''(x) > 0$$



If  $f$  is a convex function,  $0 \leq \lambda_1, \lambda_2 \leq 1$ ,  $\lambda_1 + \lambda_2 = 1$ , then for any  $x_1, x_2$ ,  $x_1 < x_2$ .

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2).$$

# Jensen's Inequality



## Theorem

Let  $X$  be a random variable. If  $g$  is a convex function, then  $E(g(X)) \geq g(E(X))$ . If  $g$  is a concave function, then  $E(g(X)) \leq g(E(X))$ . In both cases, the only way that equality can hold is if there are constants  $a$  and  $b$  such that  $g(X) = a + bX$  with probability 1.

# Quick Examples

$$\left\{ \begin{array}{l} g \text{ is convex ; } E[g(x)] \geq g[E(x)] \\ \text{Concave ; } \leq \end{array} \right.$$

1°  $g(x) = x^2, x \in \mathbb{R}$  ; Convex ;  $\frac{E[X^2] \geq (E[X])^2}{\text{Var}(X) = E[X^2] - (E[X])^2 \geq 0}$  ✓

2°  $g(x) = \frac{1}{x}, x > 0$  ; Convex ;  $E[\frac{1}{x}] \geq \frac{1}{E[x]}$  ✓

3°  $g(x) = \log x, x > 0$  ; Concave ;  $\underline{E(\log x) \leq \log(E(x))}$  ✓

# Entropy

$$E(X^2) < C$$

- Let  $X$  be a discrete r.v. whose distinct possible values are  $a_1, a_2, \dots, a_n$ , with probabilities  $p_1, p_2, \dots, p_n$  respectively (so  $p_1 + p_2 + \dots + p_n = 1$ ).
- The *entropy* of  $X$  is defined as follows:  
$$H(X) = \sum_{j=1}^n p_j \log_2 (1/p_j).$$
- Using Jensen's inequality, show that the maximum possible entropy for  $X$  is when its distribution is uniform over  $a_1, a_2, \dots, a_n$ , i.e.,  $p_j = 1/n$  for all  $j$ .
- This makes sense intuitively, since learning the value of  $X$  conveys the most information on average when  $X$  is equally likely to take any of its values, and the least possible information if  $X$  is a constant.



# Proof

① Construct a r.v.  $Y$  s.t.

$$Y = \begin{cases} \frac{1}{p_1} & \text{w.p. } p_1 \\ \frac{1}{p_2} & \text{w.p. } p_2 \\ \vdots & \vdots \\ \frac{1}{p_n} & \text{w.p. } p_n \end{cases}$$

$$E(Y) = \frac{1}{p_1} \cdot p_1 + \frac{1}{p_2} \cdot p_2 + \dots + \frac{1}{p_n} \cdot p_n = \underline{n}$$

$$\textcircled{2} \quad H(X) = \sum_{j=1}^n p_j \log_2 \frac{1}{p_j} = \underline{E[\log_2 Y]} \leq \log_2(E[Y]) = \log_2 n$$

$$\forall (p_1, \dots, p_n) \quad H(X) \leq \log_2 n \quad \Rightarrow \quad \max_{p_1, \dots, p_n} H(X) \leq \log_2 n$$

$$\textcircled{3} \quad \max_{p_1, \dots, p_n} H(X) \geq \log_2 n \quad \left[ X \sim \text{Unif}\left(\frac{1}{n}\right), p_1 = p_2 = \dots = \frac{1}{n} \right. \\ \left. H(X) = \log_2 n \right]$$

$$\textcircled{4} \quad \Rightarrow \quad \max_{p_1, \dots, p_n} H(X) = \log_2 n, \quad X^* \sim \text{Unif}\left(\frac{1}{n}\right)$$

# Kullback-Leibler Divergence

Relative Entropy

Let  $\mathbf{p} = (p_1, \dots, p_n)$  and  $\mathbf{r} = (r_1, \dots, r_n)$  be two probability vectors (so each is nonnegative and sums to 1). Think of each as a possible PMF for a random variable whose support consists of  $n$  distinct values.

The *Kullback-Leibler* divergence between  $\mathbf{p}$  and  $\mathbf{r}$  is defined as

$$D(\mathbf{p}, \mathbf{r}) = \sum_{j=1}^n p_j \log_2(1/r_j) - \sum_{j=1}^n p_j \log_2(1/p_j).$$

Show that the Kullback-Leibler divergence is nonnegative.

Proof ①  $D(p, r) = \sum_{j=1}^n p_j \log_2 \frac{1}{p_j} - \sum_{j=1}^n p_j \log_2 \frac{1}{r_j}$

$$= \sum_{j=1}^n p_j \log_2 \frac{p_j}{r_j} = - \underbrace{\sum_{j=1}^n p_j \log_2 \frac{r_j}{p_j}}.$$

② Construct a r.v.  $Y$  s.t.

$$P(Y = \frac{r_j}{p_j}) = p_j, \quad j=1, 2, \dots, n.$$

$$\Rightarrow E(Y) = \sum_{j=1}^n \frac{r_j}{p_j} \cdot p_j = \sum_{j=1}^n r_j = \underline{1}.$$

$$\begin{aligned} \textcircled{3} \quad \underline{D(p, r)} &= - \sum_{j=1}^n p_j \log_2 \frac{r_j}{p_j} = - \underline{E[\log_2 Y]} \\ &\geq -\log_2(E(Y)) = -\log_2 1 = 0 \end{aligned}$$

# Markov's Inequality

$$\underline{f(x) \leq a}$$

✓

tighter  
 $a < b$

$$\underline{f(x) \leq b}$$

✗

Concentration Inequality

Chebyshev

Markov

Lyapunov

## Theorem

For any r.v.  $X$  and constant  $a > 0$ ,

$$P(\underline{|X| \geq a}) \leq \frac{E|X|}{a}.$$

# Proof

$$P(|X| \geq a) \leq \frac{1}{a} E|X|, \quad a > 0$$

①  $Y = \frac{|X|}{a} \geq 0$ ;  $\rightarrow$  Indicator of " $Y \geq 1$ ".

$$\underline{I(Y \geq 1) \leq Y}$$

if  $Y \geq 1$  LHS RHS  $1 \leq Y$  ✓  
if  $0 \leq Y < 1$  LHS RHS  $0 \leq Y$  ✓

$$\Rightarrow E[I(Y \geq 1)] \leq E[Y]$$

$$\Rightarrow P(Y \geq 1) \leq \underline{E[Y]} = \frac{1}{a} E|X|$$

$$\Downarrow$$
$$P\left(\frac{|X|}{a} \geq 1\right)$$

$$\Downarrow$$
$$P(|X| \geq a)$$

②

$$P(|X| \geq a) \leq \frac{1}{a} E|X|$$

# Chebyshev's Inequality

$$\textcircled{1} \quad P(\underline{|X-\mu| \geq a}) = P(|X-\mu|^2 \geq a^2)$$

Markov's Inequality

$$\leq \frac{1}{a^2} E(|X-\mu|^2) = \frac{1}{a^2} \sigma^2$$

## Theorem

Let  $X$  have mean  $\mu$  and variance  $\sigma^2$ . Then for any  $a > 0$ ,

$$P(\underline{|X-\mu| \geq a}) \leq \frac{\sigma^2}{a^2} \quad \uparrow a \quad \downarrow \left(\frac{1}{a^2} \sigma^2\right)$$

# Proof

Application :

$\bar{X}_n$  : sample mean ;  $E(\bar{X}_n) = \mu$  ;  $Var(\bar{X}_n) = \frac{1}{n} \sigma^2$

$$0 \leq P(|\bar{X}_n - \mu| \geq a) \leq \frac{1}{a^2} Var(\bar{X}_n - \mu) = \frac{1}{a^2} Var(\bar{X}_n)$$

$$= \frac{\frac{1}{a^2} \cdot \frac{1}{n} \sigma^2}{\downarrow}$$

0

$$n \rightarrow (\infty)$$

$$\Rightarrow \lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq a) = 0$$

WLLN

# Chernoff's Inequality $\forall t > 0;$

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

$$\leq \frac{E[e^{tX}]}{e^{ta}}$$

## Theorem

For any r.v.  $X$  and constants  $a > 0$  and  $t > 0$ ,

$$P(X \geq a) \leq \frac{E(e^{tX})}{e^{ta}} \rightarrow \text{MGF}$$



# Proof

# Chernoff's Technique

$$(1) P(X > a) \leq \frac{E(e^{tX})}{e^{ta}} \quad \forall t > 0$$

$f(t)$

## Theorem

For any r.v.  $X$  and constants  $a$ ,

$$P(X \geq a) \leq \inf_{t > 0} \frac{E(e^{tX})}{e^{ta}}$$

$$P(X \leq a) \leq \inf_{t < 0} \frac{E(e^{tX})}{e^{ta}}$$

(1)

$$P(X \leq a), \quad \forall t < 0 \Rightarrow P(tX \geq ta)$$

$$\leq \frac{E(e^{tX})}{e^{ta}} = f(t)$$

# Proof

## Example: Normal Distribution

① MGF of  $X$ :  $M_G(X) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

②  $P(X > a) \leq \inf_{t > 0} \frac{E[e^{tx}]}{e^{ta}} = \inf_{t > 0} f(t)$

Given  $X \sim \mathcal{N}(\mu, \sigma^2)$ , for arbitrary constant  $a > \mu$ , find the Chernoff bound on  $P(X > a)$ .

③  $f(t) = \frac{1}{e^{ta}} E[e^{tx}] = \frac{1}{e^{ta}} M_G(X) = e^{\frac{\mu t + \frac{1}{2}\sigma^2 t^2 - at}{}}$

$$\frac{\mu t + \frac{1}{2}\sigma^2 t^2 - at}{=} = \frac{1}{2\sigma^2} \left[ \left( t + \frac{\mu - a}{\sigma^2} \right)^2 - \frac{(\mu - a)^2}{\sigma^4} \right]$$

$$\Rightarrow t^* = \frac{a - \mu}{\sigma^2} > 0$$

④  $P(X > a) \leq f(t^*) = e^{-\frac{(a - \mu)^2}{2\sigma^2}}$

$P(X - \mu > a - \mu) =$

# Solution

# Hoeffding Lemma

## Lemma

Let the random variable  $X$  satisfy  $\mathbb{E}(X) = 0$  and  $a \leq X \leq b$ , where  $a$  and  $b$  are constants. Then for any  $\lambda > 0$ ,

$$\mathbb{E}(e^{\lambda X}) \leq e^{\frac{1}{8}\lambda^2(b-a)^2}.$$

MGF

# Hoeffding Bound

## Theorem

Let the random variables  $X_1, X_2, \dots, X_n$  be independent with  $E(X_i) = \mu$ ,  $a \leq X_i \leq b$  for each  $i = 1, \dots, n$ , where  $a, b$  are constants. Then for any  $\epsilon \geq 0$ ,

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq \epsilon\right) \leq 2e^{-\frac{2n\epsilon^2}{(b-a)^2}}.$$

$X_n$

# Application: Parameter Estimation

$$\delta = 0.05$$

Instead of predicting a single value for the parameter, we given an interval that is likely to contain the parameter:

## Definition

$\hat{p}$  : estimation of  $p$ .

A  $1 - \delta$  confidence interval for a parameter  $p$  is an interval  $[\hat{p} - \epsilon, \hat{p} + \epsilon]$  such that

$$Pr(p \in [\hat{p} - \epsilon, \hat{p} + \epsilon]) \geq 1 - \delta.$$

$$\Leftrightarrow Pr(|\hat{p} - p| \leq \epsilon) \geq 1 - \delta$$

$$\Leftrightarrow \hat{p} - \epsilon \leq p \leq \hat{p} + \epsilon$$

$$\Leftrightarrow -\epsilon \leq p - \hat{p} \leq \epsilon$$

$$\Leftrightarrow -\epsilon \leq \hat{p} - p \leq \epsilon$$

$$\Leftrightarrow |\hat{p} - p| \leq \epsilon$$

$$\Leftrightarrow Pr(|\hat{p} - p| \geq \epsilon) \leq \delta$$



# Application: Parameter Estimation

①  $X_i \sim \text{Bern}(p)$ .  $E(X_i) = p$ ;  $0 \leq X_i \leq 1$

( $X_i = 1$  or  $0$ )

( $a=0$ ;  $b=1$ )

$E(\hat{p}) = p$ ; LLN:  $N \rightarrow \infty$ ;  $\hat{p} \rightarrow p$ .

②  $\Pr(|\hat{p} - p| \geq \varepsilon) = \Pr\left(\left|\frac{1}{N} \sum_{i=1}^N X_i - p\right| \geq \varepsilon\right)$

By Hoeffding Bound.

Tossing a coin with probability  $p$  landing heads and probability  $1 - p$  landing tails.  $p$  is unknown and we need to estimate its value from experiments results. We toss such coin  $N$  times, Let  $X_i = 1$  if the  $i$ th result is head, otherwise 0. We estimate  $p$  by using  $\hat{p} = \frac{X_1 + \dots + X_N}{N}$ . Find the confidence interval for  $p$ .

$$\leq 2e^{-\frac{2N\varepsilon^2}{(b-a)^2}} = 2e^{-2N\varepsilon^2}$$

③  $2e^{-2N\varepsilon^2} = \delta \Rightarrow \varepsilon = \sqrt{\frac{\ln(\frac{2}{\delta})}{2N}}$

## Solution

$$\textcircled{4} \quad \Pr(|\hat{p} - p| \geq \varepsilon) \leq \delta$$

$$\Rightarrow \Pr(|\hat{p} - p| < \varepsilon) > 1 - \delta$$

$$\Rightarrow \Pr(p \in (\hat{p} - \varepsilon, \hat{p} + \varepsilon)) > 1 - \delta$$

$$\textcircled{5} \quad (\hat{p} - \varepsilon, \hat{p} + \varepsilon)$$

$$\left( \frac{1}{N} \sum_{i=1}^N X_i - \sqrt{\frac{\ln(\frac{2}{\delta})}{2N}}, \frac{1}{N} \sum_{i=1}^N X_i + \sqrt{\frac{\ln(\frac{2}{\delta})}{2N}} \right)$$

1 -  $\delta$  Confidence Interval.

# Application Example: Monte Carlo Method for Estimation $\pi$

① Square  $[-1, 1] \times [-1, 1]$

Circle :  $\{(x, y) : x^2 + y^2 \leq 1\}$

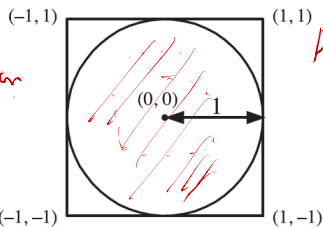
②

$$\pi = 4 \Pr(A)$$

$$= 4 \cdot E(I_A)$$

$$\approx 4 \left( \frac{1}{n} \sum_{i=1}^n z_i \right)$$

→ Indicator



$A = \text{"point landing in the circle"}$   
 $\Pr(A)$

$$= \frac{\text{area (circle)}}{\text{area (square)}}$$

$$= \frac{\pi}{4}$$

- A point chosen uniformly at random in the square has probability  $\pi/4$  of landing in the circle

③  $z_i = I_{\{\text{the } i\text{th point landing in the circle}\}}$

# Example: Monte Carlo Method for Estimation $\pi$

$$P(Z_i=1) = \frac{\pi}{4}, \quad E(Z_i) = \frac{\pi}{4}, \quad \underline{0 \leq Z_i \leq 1} \quad \underline{Z_i = 1 \text{ or } 0}$$

④ Suppose we generate  $n$  points.

$$W = \frac{1}{n} \sum_{i=1}^n Z_i \quad E(W) = \frac{\pi}{4}.$$

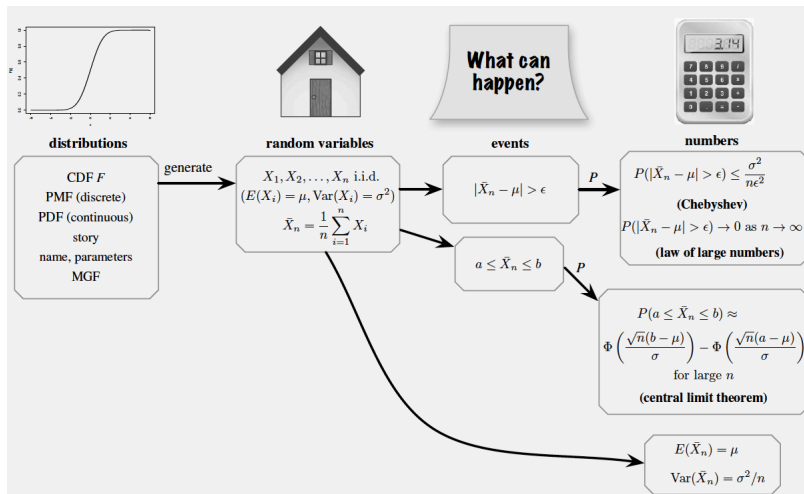
⑤  $\hat{\pi} = 4W$ .  
 $n \rightarrow \infty, \hat{\pi} \rightarrow \pi$   
Finite  $n$ :

$$\begin{aligned} \Pr(|\hat{\pi} - \pi| \geq \varepsilon) &= \Pr(|4W - \pi| \geq \varepsilon) \\ &= \Pr\left(|W - \frac{\pi}{4}| \geq \frac{\varepsilon}{4}\right) = \Pr\left|\frac{1}{n} \sum_{i=1}^n Z_i - \frac{\pi}{4}\right| \geq \frac{\varepsilon}{4} \\ &\stackrel{\text{Hoeffding Bound}}{\leq} 2e^{-\frac{2n(\frac{\varepsilon}{4})^2}{(1-0)^2}} = \underline{2e^{-\frac{1}{8}n\varepsilon^2}} = \delta \end{aligned}$$

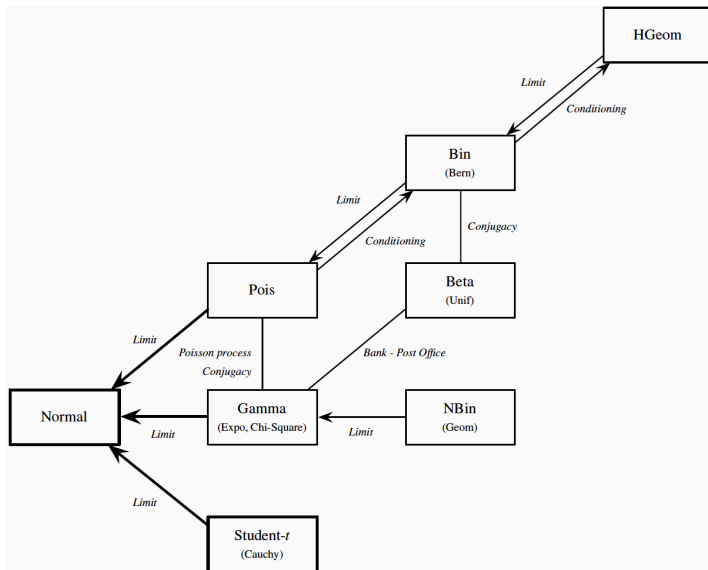
⑥  $\varepsilon = \sqrt{\frac{8 \ln(\frac{2}{\delta})}{n}} \Rightarrow \Pr(\pi \in (\hat{\pi} - \sqrt{\frac{8 \ln(\frac{2}{\delta})}{n}}, \hat{\pi} + \sqrt{\frac{8 \ln(\frac{2}{\delta})}{n}})) > 1 - \delta$

# Example: Monte Carlo Method for Estimation $\pi$

# Summary 1



# Summary 2



# References

- Chapter 10 of **BH**
- Chapter 5 of **BT**